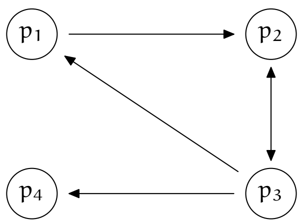
*Topic*

# Page Ranking

Imagine that you are looking for the best book on Linear Algebra. You probably would try a web search engine such as Google. These lists pages ranked by importance. The ranking is defined, as Google's founders have said in [Brin & Page], that a page is important if other important pages link to it: "a page can have a high PageRank if there are many pages that point to it, or if there are some pages that point to it and have a high PageRank." But isn't that circular — how can they tell whether a page is important without first deciding on the important pages? The answer is to use eigenvalues and eigenvectors.

We will present a simplified version of the Page Rank algorithm. For that we will model the World Wide Web as a collection of pages connected by links. This diagram, from [Wills], shows the pages as circles and the links as arrows. Page $p_1$ has a link to page $p_2$. Page $p_2$ has a link to $p_3$. And $p_3$ has links to $p_1$, $p_2$, and $p_3$




The key idea is that pages that should be highly ranked if they are cited often by other pages. That is, we raise the importance of a page $p_i$ if it is linked-to from page $p_j$. The increment depends on the importance of the linking page $p_j$ divided by how many out-links $a_j$ are on that page.

$$\mathcal{I}(p_i) = \sum_{\text{in-linking pages } p_j} \frac{\mathcal{I}(p_j)}{a_j}$$

Thus, the importance of $p_1$ equals $1/3$ times the importance of $p_3$, since the only link to $p_1$ comes from $p_3$. Similarly the importance of $p_2$ is the sum of the importance of $p_1$ plus $1/3$ times the importance of $p_3$.

This stores the information.

$$\begin{pmatrix} 0 & 0 & 1/3 & 0 \\ 1 & 0 & 1/3 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1/3 & 0 \end{pmatrix}$$

The algorithm's inventors describe a way to think about that matrix.

> PageRank can be thought of as a model of user behavior. We assume there is a 'random surfer' who is given a web page at random and keeps clicking on links, never hitting "back" ... The probability that the random surfer visits a page is its PageRank. [Brin & Page]

Thus, looking at the first row of the matrix, the only way for a random surfer to get to $p_1$ is to have come from $p_3$. That page has three links so the chance of clicking on $p_1$'s is $1/3$ times the chance of being on page $p_3$.

This brings up the question of page $p_4$. On the Internet many pages are *dangling* or *sink links*, without any outbound links. What happens to the random surfer who visits this page? The simplest thing is to imagine that the surfer chooses the next page entirely at random.

$$H = \begin{pmatrix} 0 & 0 & 1/3 & 1/4 \\ 1 & 0 & 1/3 & 1/4 \\ 0 & 1 & 0 & 1/4 \\ 0 & 0 & 1/3 & 1/4 \end{pmatrix}$$

We will find vector $\vec{\mathcal{I}}$ whose components are the importance rankings of each page $\mathcal{I}(p_i)$. With this notation, our requirements for the page rank are that $H\vec{\mathcal{I}} = \vec{\mathcal{I}}$. That is, we want an eigenvector of the matrix associated with the eigenvalue $\lambda = 1$.

Here is *Sage*'s calculation of the eigenvectors (edited to fit the page).

```
sage: H=matrix([[0,0,1/3,1/4], [1,0,1/3,1/4], [0,1,0,1/4], [0,0,1/3,1/4]])
sage: H.eigenvectors_right()
[(1, [(1, 2, 9/4, 1)], 1),
 (0, [(0, 1, 3, -4)], 1),
 (-0.3750000000000000? - 0.4389855730355308?*I,
    [(1, -0.1250000000000000? + 1.316956719106593?*I,
         -1.875000000000000? - 1.316956719106593?*I, 1)], 1),
 (-0.3750000000000000? + 0.4389855730355308?*I,
    [(1, -0.1250000000000000? - 1.316956719106593?*I,
         -1.875000000000000? + 1.316956719106593?*I, 1)], 1)]
```

The eigenvector that *Sage* gives associated with the eigenvalue $\lambda = 1$ is this.

$$\begin{pmatrix} 1 \\ 2 \\ 9/4 \\ 1 \end{pmatrix}$$

Of course, there are many vectors in that eigenspace. To get a page rank number we normalize to length one.

```
sage: v=vector([1, 2, 9/4, 1])
sage: v/v.norm()
(4/177*sqrt(177), 8/177*sqrt(177), 3/59*sqrt(177), 4/177*sqrt(177))
sage: w=v/v.norm()
sage: w.n()
(0.300658411201132, 0.601316822402263, 0.676481425202546, 0.300658411201132)
```

So we rank the first and fourth pages as of equal importance. We rank the second and third pages as much more important than those, and about equal in importance as each other.

We'll add one more refinement. We will allow the surfer to pick a new page at random even if they are not on a dangling page. In this equation it happens with probability $1 - \alpha$.

$$G = \alpha \cdot \begin{pmatrix} 0 & 0 & 1/3 & 1/4 \\ 1 & 0 & 1/3 & 1/4 \\ 0 & 1 & 0 & 1/4 \\ 0 & 0 & 1/3 & 1/4 \end{pmatrix} + (1 - \alpha) \cdot \begin{pmatrix} 1/4 & 1/4 & 1/4 & 1/4 \\ 1/4 & 1/4 & 1/4 & 1/4 \\ 1/4 & 1/4 & 1/4 & 1/4 \\ 1/4 & 1/4 & 1/4 & 1/4 \end{pmatrix}$$

This is the *Google matrix*.

In practice $\alpha$ is typically between 0.85 and 0.99. Here are the ranks for the four pages with various $\alpha$'s.

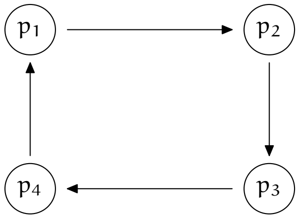
| $\alpha$ | 0.85 | 0.90 | 0.95 | 0.99 |
|---|---|---|---|---|
| $p_1$ | 0.325 | 0.317 | 0.309 | 0.302 |
| $p_2$ | 0.602 | 0.602 | 0.602 | 0.601 |
| $p_3$ | 0.652 | 0.661 | 0.669 | 0.675 |
| $p_4$ | 0.325 | 0.317 | 0.309 | 0.302 |

The details of the algorithms used by commercial search engines are secret, have many refinements, and also change frequently. But the inventors of Google were gracious enough to outline the basis for their work in [Brin & Page]. A more current source is [Wikipedia, Google Page Rank]. Two additional excellent expositions are [Wills] and [Austin].

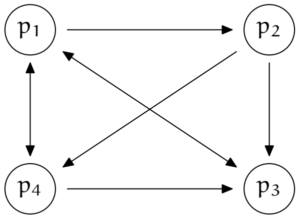
## Exercises

1 A square matrix is *stochastic* if the sum of the entries in each column is one. The Google matrix is computed by taking a combination $G = \alpha * H + (1 - \alpha) * S$ of two stochastic matrices. Show that G must be stochastic.

2 For this web of pages, the importance of each page should be equal. Verify it for $\alpha = 0.85$.




3 [Bryan & Leise] Give the importance ranking for this web of pages.




**(a)** Use $\alpha = 0.85$.

**(b)** Use $\alpha = 0.95$.

**(c)** Observe that while $p_3$ is linked-to from all other pages, and therefore seems important, it is not the highest ranked page. What is the highest ranked page? Explain.